Assignment 2

Doornakkers-West: Idea evaluation through prototyping and testing

Filippos Karakaxas - 1950835
Pangeran Tambunan - 2001063
Raafi Sedyaporna - 1923250
Marc Boglioni - 1818848
Dimitrios Tsioutsas - 1786458

Group 26



Content *overview*

1

Problems Addressed

2

Prototypes of Ideas

3

Concept Testing Results

4

Ideas Assessment via MON

5

Ethical Consideration

6

Final Idea selection

Problems Addressed

Summary of Doornakkers-West data

Doornakkers-West, Eindhoven is a dense, young-adult, highly mixed, rental neighbourhood

- 3,465 residents at 4,900 per km² — roughly nine times Dutch average density
- 53% of households are one person living alone; 33% of residents are 25–45, 14% are 65+
- Origin: 52% Netherlands, 16% other European, 33% outside Europe
- 56% social rent; housing stock predominantly built 1925–1950
- Average income €30,700 per resident; ~10% of households at or near the social minimum; 68% labour participation
- 165 recorded offences and 16 traffic incidents in 2025

Numbers do not explain: on the municipality's own wellbeing survey the area scores poorly on happiness and on participation in clubs and sport, while residents rate their own physical health as fine. The gap is social, not medical — which is what sent us into the neighbourhood to ask why.

Summary of interviews

1. Nothing organised at neighbourhood level. Everyone says it; "No association, club, group, team or organisation in this neighbourhood. The few social places that exist are not open to everyone.
2. Social life is segmented by gender, age and background. "They're all old men, I would not go there." On the many barbershops and bars: she would not walk in. Residents go elsewhere to socialise: Tongelrestraat cafés, the centrum, "another city".
3. Evening and night are when people feel unsafe — and that is when they would meet. A 16-year-old carries a knife after a child was kidnapped two years ago
4. The street discourages being outside. Speeding and aggressive driving on the main roads; one road described as "terrible", with promised repairs that never started If you arrive, you cannot get in.
5. Some people don’t know anyone here by name. Says hello, small talk, nothing more. Language is named repeatedly as a barrier, by Dutch and non-Dutch residents alike

Problem statement + addressed pains

PROBLEM STATEMENT: Residents of Doornakkers-West across a wide range of ages want to be active outdoors and get to know their neighbours, but in the evening and at night, parks and other social spaces feel unsafe and offer few natural opportunities to meet. This keeps people in private circles, so neighbours stay strangers, safety concerns go unshared, and the everyday contact that makes a place feel safe erodes.**

Personas: Jan (19) — carries a knife so he can keep enjoying the outdoor life he loves.

Maria (43) — deep roots here, yet takes her social life to the centrum, because there is no shared place to meet.

How might we make Doornakkers-West's evening public spaces feel safe and inviting enough for neighbours of all ages to use and share?

Prototypes of Ideas

Prototype summaries

For the development of the prototypes, we relied on the conclusions and insights from our interviews. Our ideation of the solutions in the workshops was a result of brainstorming in the workshop sessions, from which we developed prototypes. These prototypes were created through photos, sketches, and more. These prototypes also will be explained in depth in later slides.

- Social cafe -> prototype of an app
  - The missing neighbourhood-level layer and the missing shared place — one room that belongs to everyone, open in the evening, where joining something does not require already knowing someone
- Security service -> prototype of a poster
  - People feel unsafe, especially at evenings/nights which is a time where people do often want to meet and have events. A communal security service fosters togetherness and increases safety
- Non-Newtonian speedbump -> prototype of a 3D render
  - The street that pushes people indoors — speeding and aggressive driving near parks, schools, playgrounds and community facilities, which residents named as their main safety complaint

Each prototype is evaluated in the following sections on desirability (concept testing), feasibility and viability (MON), and ethics (Ethical Cycle).

Prototype 1: Social cafe

Mobile application/web UI



Source: https://brabantmaatjes.nl/social-cafe-tilburg/

Brief prototype description

- App (full web features included) of a social cafe
- Cafe doubles down as large recreational area, open to all demographics with activities
- Application is constantly updated. Feedback often provided to make UI look better

Features from app:

Homepage	Hub to reach other parts of app
Activities dashboard	Overview of upcoming activities
Calendar + booking	Calendar of all events to book
Cafe menu	Menu of the central cafe that connects all rooms
Gallery	Pictures of all different events in archive
“Make your own event”	Feature for ANYONE to make their own event
Clubs and community	General page for all available clubs & descriptions

When and where would it be used?

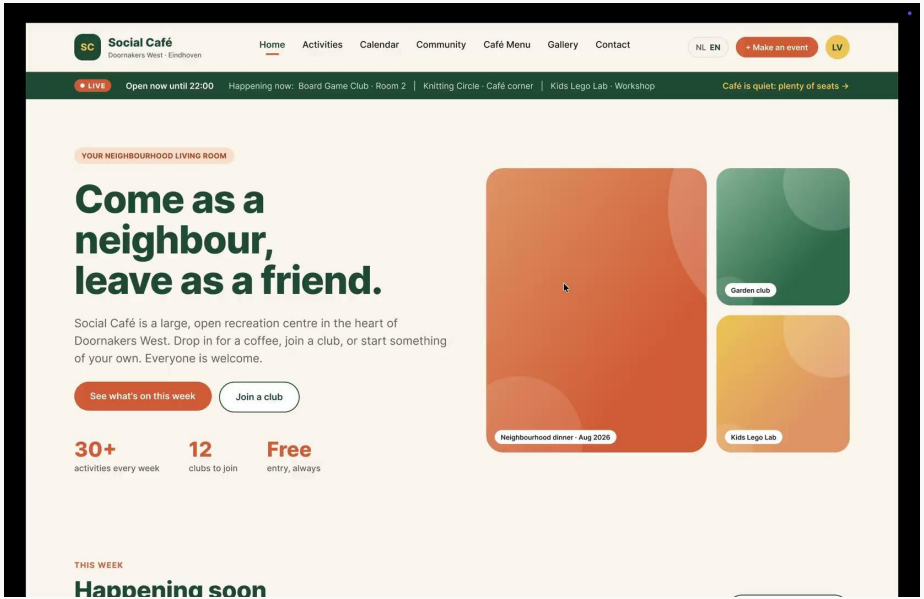
- Weekday evenings and weekends, when nothing else in the neighbourhood is open and residents told us they travel to Tongelrestraat or the centrum instead.
- A ground-floor unit on a route people already walk, with lit windows facing the street — close to the park and football pitch rather than the canal edge.
- Year-round, covering October to April, when the "grandpa playground" that residents call the most social place in the neighbourhood is closed.

What problem does it solve?

- Every resident we interviewed independently said the same thing: there is no association, club or organisation at neighbourhood level, "it's hard to do activities with people." Social life therefore runs only on ties people already have — a resident of one year still knows nobody by name, and the venues that do exist are segmented by age and gender. This concept supplies the missing shared layer: one place that belongs to everyone, open in the evening, where starting or joining an activity does not require already knowing someone. By giving residents a lit destination after dark, it also acts on the loop in our problem statement, where empty evening streets and unfamiliar neighbours reinforce one another.

Prototype design

We have an explanatory video explaining the app. It can be viewed below embedded in the slides, or available in the following link: https://drive.google.com/file/d/1rPsiW51kxL3GlKLKhsEEaOFnQ9WsVy5o/view?usp=sharing



Prototype 2: Security Service

Advertising poster



Source: https://globalsecuritysolutionsllc.com/improving-neighborhood-security/

Brief prototype description

- Neighbourhood evening patrol service for Doornakkers-West, advertised to residents by poster.
- Two uniformed, licensed stewards on marked patrol, 19:00–06:00, every night.
- Poster is bilingual (Dutch and English), designed for shops, schools and housing-block entrances. Residents report by phone or QR code; 112 stays the route for emergencies.
- Not enforcement — stewards observe, assist and report, and know when to call the police.

Poster includes:

Poster element	Purpose
Headline "Safe after dark"	One benefit, one line, readable at a distance
Hours 19:00–06:00	Names the exact window residents avoid
Marked car and uniformed guard	Makes the service recognisable before it
Phone number & QR Code	Reporting route for people either online or offline

When and where would it be used?

- Every evening and night, 19:00–06:00 — the hours residents named as the ones they avoid being outside.
- On the routes and places residents said they avoid after dark: the canal path, the park edges, side streets and the walk home.
- Places where traditional speed bumps cause unnecessary discomfort for drivers going at safe speeds.
- Posters where residents already go — local shops, schools, the supermarket and housing-block entrances — so the service is known before it is needed.

What problem does it solve?

- Residents say evening is both when they would most like to be outside and when they feel least able to be. One 16-year-old carries a knife after a child was kidnapped in the neighbourhood two years ago; a parent said they feel "a little bit more uncomfortable walking in the dark". Because the evening is also when people would otherwise meet, that fear removes the very hours in which neighbourly contact could happen. This concept places a visible, approachable presence on the street during exactly those hours and gives residents a simple way to report what they see, so that using the neighbourhood after dark no longer depends on how brave someone feels.

Prototype design



Fig 1. Chosen poster template





Figs 2 & 3. Rejected ideas of poster

Figs 2 & 3 were rejected due to not having eye-catching designs and not really fitting the the theme of emphasizing the night safety, bad color choices, consensus of the group, and opinions of people interviewed later.

Prototype 3: Non-Newtonian speed bump



Computer-generated render

Brief prototype description

- Adaptive speed bump containing a sealed chamber filled with oobleck.
- Oobleck: Mixture of cornstarch and water.
- The chamber sits inside a durable, rubber casing with the usual reflective road markings.

Oobleck Response:

Low speed	Pressure builds slowly, allowing the mixture to **move** and the bump to **compress**, allowing drivers within the speed limit to cross **smooth**.
High speed	Quick impact causes the oobleck to temporarily **resist deformation**. The bump feels **harder** and produces a stronger **slowing effect**.
After impact	Material returns to its **fluid-like state** and bump regains its **normal shape**.

When and where would it be used?

- Residential streets and intersections with many speeding complaints.
- Near parks, schools, playgrounds and community facilities.
- Places where traditional speed bumps cause unnecessary discomfort for drivers going at safe speeds.

What problem does it solve?

- Based on the initial interviews that were conducted, residents had expressed their safety concerts regarding uncontrolled intersection and aggressive driving. This concept aims to create a speed-dependent response, rewarding careful drivers with a smoother crossing, but providing a high resistance to vehicles travelling at speeds that surpass the speed limit.

Prototype Designs

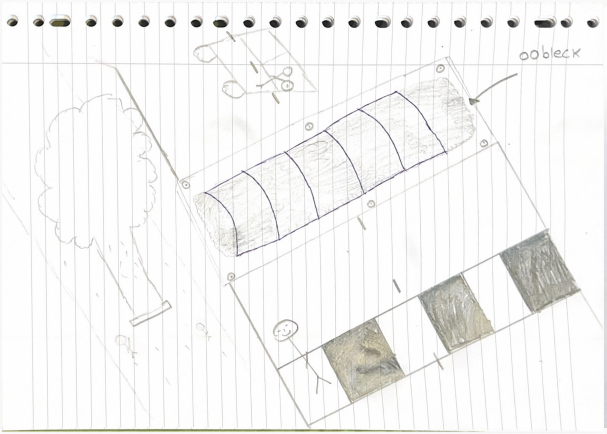


Fig 1. Ideation Sketch



Fig 2. AI Render of Product



Fig 3. Implementation of Product via AI

Concept Testing Results

CONCEPT TESTING RESULTS

Non-Newtonian speed bump Prototype

Remembered:

- It looks like a regular black-and-yellow bump, with only the cutaway hinting at the special filling.
- "Soft when slow, hard when fast" was clear only after explanation.
- The street image felt realistic and recognisable.

Found desirable:

- Only fast drivers are affected, so slow drivers are not.
- All six participants saw it as relevant near the park, school and supermarket crossing, helping children, older people and cyclists.
- All participants would support it.

Unclear:

- The hard state and how hard it gets are not shown, and the effect on bikes, wheelchairs, rollators, emergency vehicles and tripping risk is open.
- Durability, noise, cost and maintenance are open questions, as is whether it is real or tested, since the images are AI-made.
- It looks low and narrow, so a car might drive around it, and it addresses traffic but not evening comfort.

Suggestions:

- Show a video or animation of a car and a bike going over it, slow and fast, plus an example or test results from another city.
- Pilot it on one street near a school or park and measure speeds before and after.
- Combine it with a marked low-curb crossing, signage, better lighting and a check of traffic-light timing, and show people with a cane or rollator in the visuals.

CONCEPT TESTING RESULTS

Security Service Prototype

Remembered:

- The headline "Safe after dark" and the hours 19:00–06:00.
- Marked patrol cars with uniformed guards.
- The ways to report (phone and QR code) and the 112 reminder.

Found desirable:

- All six would feel at least somewhat more at ease after dark, most for evening routes home.
- Trained guards who know when to call the police, the "for every resident" message and the bilingual text.
- Easy reporting, a simple poster, and mostly positive support (one respondent would put the other two ideas first).

Unclear:

- The phone number and QR code are placeholders, and it is unclear who the guards or company are, who pays and how they work with the police.
- Response time, privacy (for example, filming) and how guards treat the people they meet are not explained.
- It is only in Dutch and English, 19:00–06:00 every night seems long, and a patrol cannot be everywhere.

Suggestions:

- Fill in the phone number and QR code, add more languages, and show guards on foot talking to residents.
- Hire people from the neighbourhood, publish clear rules and a complaint option, and add a patrol-route map including the canal and side streets.
- Offer a "call and be met on my route" option, hang the poster in local shops and schools, and combine the patrol with good street lighting.

CONCEPT TESTING RESULTS

Social Cafe Prototype

Remembered:

- The calendar with booking and reminders, free entry, and that anyone can drop in.
- Low prices, "pay what you can" and "Buy a coffee for a neighbour".
- Clubs with named hosts, the noticeboard, "Make your own event" and opening hours until 22:00.

Found desirable:

- A place nearby to meet neighbours and friends, with activities across ages such as the kids' club, family games, chair yoga and language cafés.
- Low prices, a varied menu with halal and vegan labels, the Welcome host role and the accessibility box in the booking form.
- All six participants would support it.

Unclear:

- Only a desktop layout was shown, it is only in Dutch and English, and people who are not online might be left out.
- The address and phone number are placeholders, and practical details are missing (step-free entrance, benches, accessible toilet, wifi, route).
- The indoor kids' area is hard to see, there is little for teenagers or sports, and it is unclear who moderates the noticeboard and how noise for nearby homes is handled.

Suggestions:

- Add a mobile version and more languages, offer phone booking, and put printed calendars in schools and local shops.
- Add a youth and sports category, and add accessibility details with entrance photos, a map and route, and wifi and study information.
- Add a welcome event for newcomers, show the kids' area, and keep offline information available.

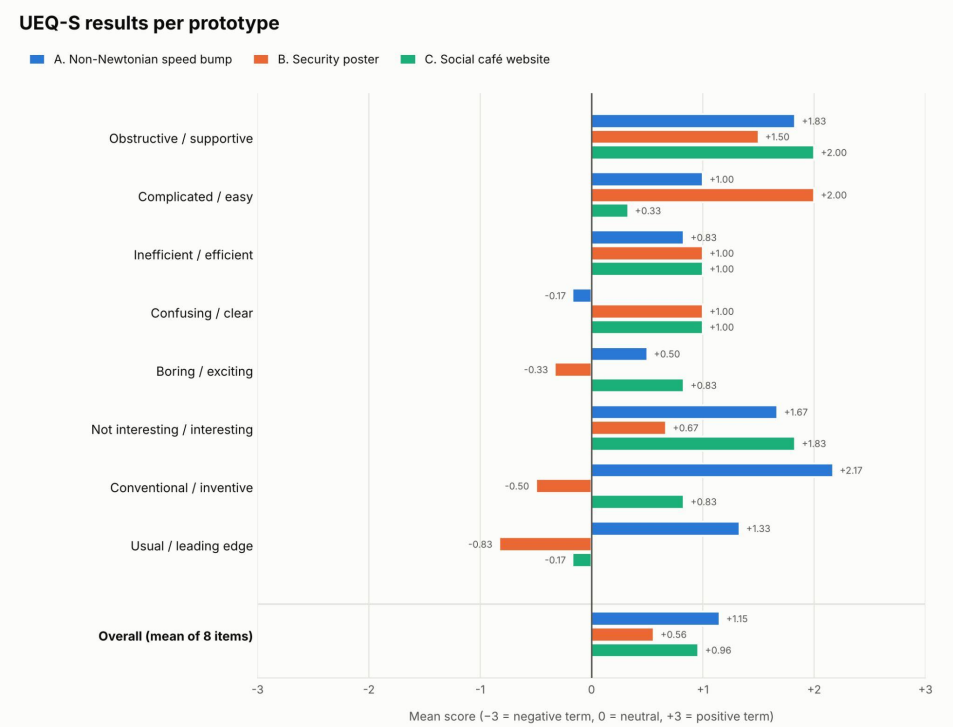
CONCEPT TESTING RESULTS

How the UEQ-S scores were calculated:

Each of the six participants ticked one of 7 circles per item, running from the negative word (left) to the positive word (right). Each tick was converted to a score from −3 to +3, so the middle circle is 0. The item score is the mean across the six respondents. The overall score is the mean of all eight item scores.

Item	A. Speed bump	B. Security poster	C. Café website
Obstructive / supportive	+1.83	+1.50	+2.00
Complicated / easy	+1.00	+2.00	+0.33
Inefficient / efficient	+0.83	+1.00	+1.00
Confusing / clear	−0.17	+1.00	+1.00
Boring / exciting	+0.50	−0.33	+0.83
Not interesting / interesting	+1.67	+0.67	+1.83
Conventional / inventive	+2.17	−0.50	+0.83
Usual / leading edge	+1.33	−0.83	−0.17
Overall (mean of 8 items)	+1.15	+0.56	+0.96

CONCEPT TESTING RESULTS



Ideas Assessment (MON)

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

POTENTIAL

COMPELLING REASON TO BUY:

UNMET NEEDS:

- Residents have limited contact outside existing social circles.
- Some residents avoid public spaces in the evenings.
- Participation in associations and clubs is low.
- Financial concerns could potentially limit access to social activities.
- Residents are keen for more natural and informal ways to meet.

EFFECTIVE SOLUTIONS:

- Regular reason to visit the park.
- Informal contact rather than a forced participation.
- Activities for different age groups.
- Neighbourhood centre point for information and local initiatives.

BETTER THAN ALTERNATIVES:

- Located where residents already walk and do sports.
- Low-cost products.
- Huge variety of sports available.
- Separate areas within the café for different purposes (etc. board game area).
- Alcohol-free café.
- Community events organised.

Rating: High

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

MARKET VOLUME:

CURRENT MARKET SIZE:

- The café's primary market consists of residents living close enough to visit regularly.
- Doornakkers-West => 3,465 residents [1].
- Entire Doornakkers district => 8,783 residents [2].

Assumption: "10% of the residents become regular customers and each person makes two 5 euros purchases."

Doornakkers-West => 347 regular customers:

347*(2*5) = 3,470 euros monthly revenue.

41,640 euros annual revenue.

Entire Doornakkers district => 878 regular customers:

878*(2*5) = 8,780 euros monthly revenue.

105,360 euros annual revenue.

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

MARKET VOLUME:

EXPECTED MARKET GROWTH:

Population growth: Eindhoven's population is expected to increase to 305,000 in 2040(25% increase from 2024), but there is still no evidence that Doornakkers-West will grow at the same rate [3].

Growing food-and-drinks market: The Dutch food-and-drinks establishments recorded 3.5% year-on-year turnover growth in 2026, and cafés recorded 3.2% growth. Therefore, this demonstrates a modest growth although this could mainly be due to increasing prices rather than higher number of customers [4].

Geographical expansion: The most reasonable and realistic growth in market route is expanding from just Doornakkers-West to the entirety of its district. As previously indicated, this would increase the target population from 3,465 to 8,783, and would thus generate more revenue.

Rating: Mid

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

ECONOMIC VIABILITY:

In order to assess the economic viability the moderate assumptions below will be used:

- Opening: 5 days per week
- Opening time: 6 hours per day
- Average customer spending: 5 euros
- Food, drinks and packaging costs: 35% of food-and-drinks sales
- Paying association members: 100
- Association membership fee: 5 euros per month
- Facility bookings: 8 per month
- Average booking duration: 3 hour
- Facility rental price: 25 euros per hour
- Hospitality labour cost: 27 euros per hour
- Other fixed operating costs: 1,200 per month

Food & Drinks Revenue:

Assumption: “60 customers per day.”

60 customers *5 euros*(5 days*4 weeks) = 6,000 euros clean revenue per month.

6,000 euros * 35% = 2,100 euros cost per month.

6,000 euros - 2,100 euros = 3,900 euros revenue per month (not profit yet as staff and other fixed costs need to get payed).

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

ECONOMIC VIABILITY:

Association Membership Revenue:

100 members * 5 euros = 500 euros clean revenue.
Assumption: " 20% of this cost is used for administration, activity materials and other relevant costs."
500 euros * 20% = 100 euros cost per month.
500 euros - 100 euros = 400 euros revenue per month.

Facility-Rental Revenue:

8 bookings per month * 3 hour duration * 25 euros per hour = 600 euros per month clean revenue.
Assumption: " 15% is required for additional cleaning, electricity and preparation."
600 euros * 15% = 90 euros costs.
600 euros - 90 euros = 510 euros revenue per month.

Total Contribution Revenue:

3,900 euros + 400 euros + 510 euros = 4,810 euros total revenue per month.

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

ECONOMIC VIABILITY:

Fixed Monthly Costs:

5 days * 4 weeks = 20 days per month.

20 days * 6 hours = 120 working hours per month.

120 hours * 27 euros = 3,240 euros cost

3,240 euros cost + 1,200 euros other fixed operating costs = 4,440 euros total fixed costs.

Approximate Monthly Operating Profit:

4,810 euros - 4,440 euros = 370 euros profit.

- The profit margin is relatively small, however this assumption is just for a very moderate scenario.
- Via other more extreme scenarios a high amount of profit could be generated.

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

CHALLENGE

Customers' ability to pay:

- Majority of residents can afford the membership subscription of 5 euros per month.
- Membership is reduced or free for residents of a low-income bracket.
- Residents do not need to pay to sit at the café.
- Local organisations or municipality can sponsor memberships.
- Facility rental prices are going to differ for community residents and commercial organisations, to allow community members to access them at a lower cost.

Customer Stickiness:

Promotion through:

- Monthly memberships
- Regular association meetings
- Recurring facility bookings
- Loyalty cards
- Weekly events and activities
- Discount for members
- Emotional attachment

Rating: High

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

CHALLENGE

IMPLEMENTATION OBSTACLES:

- Catering operating permit
- Compatibility with the municipal environment plan
- Public space permission from municipality or private owner
- NVWA food-business registration
- HACCP-based hygiene procedures
- Insurance
- Waste management
- Access to electricity, water and storage
- Construction time could take long
- Funding and investment risks since the neighbourhood has never had a business as such

Rating: Mid

TIME TO REVENUE:

Stages:

- Café tent/table => 2-4 weeks
- Site permission and permanent café place => 1-3 months
- Operate a regular pop-up => 2-4 months
- Build and open the actual full-scale social café => 6-12+ months

Terefore, time to revenue can start as soon as a tent/temporary stand is set up, however the actual revenue will start when the full-scale café is built.

Rating: Mid

IDEA ASSESSMENT (MON) - SOCIAL CAFE

Market Opportunity: Social Café

CHALLENGE

EXTERNAL RISKS:

- Rain and cold whether could prevent residents from leaving their houses and going to the café
- Residents liking the social purpose but are not willing to purchase products
- Noise complaints from events
- Vandalism or theft, especially since the neighbourhood is not very safe
- Competition from supermarkets, other cafés and existing community facilities

Rating: Mid

Overall Potential => High

Overall Challenge => Mid

IDEA ASSESSMENT (MON) – SECURITY SERVICE

Market Opportunity: Security Service

POTENTIAL

COMPELLING REASON TO BUY:

UNMET NEEDS:

- Safety concerns in parks and public spaces
- Low outdoor activity and neighbourhood social contact

EFFECTIVENESS OF SOLUTION:

- Make residents feel that someone is available nearby
- Identify broken lights, nuisance, dangerous driving or intimidation
- Creates a direct reporting way of communication
- Build a good, familiar relationship between safety workers and residents
- Increases residents' confidence in using parks and streets after dark

BETTER THAN ALTERNATIVES:

- More consistently present during evening hours
- Specifically assigned to one neighbourhood
- More approachable for residents rather than enforcement oriented services like the police

Rating: High

IDEA ASSESSMENT (MON) - SECURITY SERVICE

Market Opportunity: Security Service

MARKET VOLUME:

Customers:

- Municipality of Eindhoven
- Housing corporations like Holland2stay with properties in Doornakkers-West
- Local business groups
- Community organisations
- Healthcare and welfare organisations

Assumption:

- One evening-security contract for Doornakkers-West
- Two safety stewards
- Six hours per evening
- 5 evenings per year
- 52 weeks per year
- 30 euros total labour cost per hour
- 21,000 euros additional security service costs per year

2 stewards * 6 hours * 5 days * 53 weeks = 3,120 hours per year
3,120 hours * 30 euros per hour = 93,000 euros direct labour cost
93,000 euros + 21,000 euros = 114,000 euros annual service cost
A contract price of approximately 150,000 euros per neighbourhood per year is therefore a good estimate, in order to generate some profit.

IDEA ASSESSMENT (MON) - SECURITY SERVICE

Market Opportunity: Security Service

CURRENT MARKET SIZE SCENARIOS:

Market Scenarios	Contracts	Potential Annual Revenue
Doornakkers-West	1	150,000 euros
Three Eindhoven neighbourhoods	3	450,000 euros
5% of Dutch municipalities	17	2.55 million euros
20% of Dutch municipalities	68	10.2 million euros
One contract per municipality	342	51.3 million euros

*Approximately 342 municipalities in the Netherlands

IDEA ASSESSMENT (MON) – SECURITY SERVICE

Market Opportunity: Security Service

EXPECTED MARKET GROWTH: [5]

- Security-sector revenue increased by approximately 1% in 2025.
- 60% of surveyed companies expected revenue growth in 2026, by approximately 9%.
- The remaining 40% expected stable revenue.

Rating: High

This is a positive evidence for the expected market growth regarding this opportunity, although the 9% figure is just an expectation rather than an actual result.

IDEA ASSESSMENT (MON) – SECURITY SERVICE

Market Opportunity: Security Service

ECONOMIC VIABILITY:

MARGINS:

Monthly labour cost:

- 2 stewards * 6 hours * 5 days * 4 weeks = 240 hours per month
- 240 hours * 30 euros per labour hour = 7,200 euros labour cost per month
- Additional monthly operating costs approximately 2,000 euros (assumption)
- 7,200 euros + 2,000 euros = 9,200 euros total estimated monthly costs

Estimated revenue and profit:

- Monthly contract payment from customer => 150,000 euros / 12 months = 12,500 euros per month
- Estimated monthly operating costs (assumption) => 10,000 euros per month
- 12,500 euros - 10,000 euros = 2,500 euros profit per month

IDEA ASSESSMENT (MON) – SECURITY SERVICE

Market Opportunity: Security Service

ECONOMIC VIABILITY:

CUSTOMER ABILITY TO PAY:

- Residents should not be the customers as even with an appropriate monthly fee the accumulated revenue would not be over the operating costs. Therefore, that would result in a loss rather than a profit.
- The main customer should be the municipality of Eindhoven. It recognises that neighbourhood safety and livability should be taken care of and should be a funding priority. However, a 150,000 euros professional contract would require a larger municipal budget [6].

CUSTOMER STICKINESS:

In general, security contracts have strong customer stickiness since:

- Safety requires continuous coverage
- Municipalities and corporations value service continuity.
- In particular, municipalities and governments are willing to stick with opportunities that improve the image of the neighbourhoods and cities.

Rating: High

IDEA ASSESSMENT (MON) - SECURITY SERVICE

Market Opportunity: Security Service

CHALLENGE

IMPLEMENTATION OBSTACLES:

- Licensing and legal compliance.
- Limited authority in public space. Stewards could only report something and cannot do something immediately.
- Camera and privacy restrictions. Monitoring public spaces needs the mayor of the municipality to justify the necessity, location and duration of public camera surveillance.
- Cooperation between stakeholders may be hard as each might have a different expectation.

Rating: High

IDEA ASSESSMENT (MON) - SECURITY SERVICE

Market Opportunity: Security Service

TIME TO REVENUE:

Stage	Estimated Time	What Occurs
Define measures	2-3 weeks	Process of selecting patrol locations, operating hours, and other performance measures
Finding a licensed security partner [7]	2-4 weeks	Potential partners can be checked through Justis's official register of Wpbr-licensed organisations.
Obtain financial support [8]	4-8 weeks	Present the idea to the municipality, police, residents and local organisations. These organisations will cooperate with each other in order to agree on the security service idea.
Contracting [9]	4-8 weeks	Customer considers the contract offer and signs it.
Operational preparation [10]	2-4 weeks	Finalise routes, reporting procedures and other important protocols. If new guards require permission and identification, the police procedure may take up to eight weeks.
First service delivered	3-5 months	First patrol shifts take place and revenue starts coming in as invoices.

Rating: High

IDEA ASSESSMENT (MON) – SECURITY SERVICE

Market Opportunity: Security Service

EXTERNAL RISKS

- Municipal budget or policy changes
- Competition from existing public services
- Public procurement and loss of bids
- Dependence on one large customer
- Security staff shortage
- High labour costs could cause a reduction in profit
- Dependence on police cooperation. If a serious incident occurs and the police is not there on time, residents may blame the service.

Rating: High

Overall Potential => High

Overall Challenge => High

IDEA ASSESSMENT (MON) - SPEED BUMP

Market Opportunity: Non-Newtonian speed bump

POTENTIAL

COMPELLING REASON TO BUY:

- **Local need:** According to BuurtKijker data collected, 38% of Doornakkers-Oost residents experience nuisance from speeding vehicles [11].
- **Strategic fit:** Eindhoven is expanding 30 km/h zones and expects approximately 70 roads to require speed reduction. However, the municipality is cautious about conventional bumps because they can obstruct emergency services [12].
- **Strong functionality:** This bump is intended to remain softer for slower vehicles but temporarily stiffen under faster impacts. Research supports this material behaviour and its potential application in speed bumps [13].

Rating: High

Current potential market: Low due to this speed bump being untested.

Rating: Super High

IDEA ASSESSMENT (MON) - SPEED BUMP

Market Opportunity: Non-Newtonian speed bump

MARKET VOLUME:

Seventy roads in Eindhoven are expected to require a lower speed limit, however not each road will require speed bumps [14].

- Assumption: " 20% of the 70 roads will require two speed bump (to cover both car currents)."
- Assumption: " Speed bump selling price of 8,000 euros."

70 roads * 20% * 2 speed bumps = 28 speed bumps

28 speed bumps * 8,000 euros = 224,000 euros total

Long-term potential market: High because municipalities in the Netherlands manage speed-reduction projects.

IDEA ASSESSMENT (MON) – SPEED BUMP

Market Opportunity: Non-Newtonian speed bump

ECONOMIC VIABILITY:

A conventional full-width modular speed bump currently costs approximately €5,761.50, without any additional taxes or installation fees [14].

Assumptions:

- Non-Newtonian speed bump is 9,000 euros
- Casing, fluid and internal cells is 2,900 euros.
- Manufacturing and quality control is 800 euros.
- Transport and Installation is 1,500 euros.
- Warranty reserve is 500 euros.

Total estimated cost: 2,900 + 800 + 1,500 + 500 = 5,700 euros

Profit: 9,000 - 5,700 = 3,300 euros

Rating: Mid

(Just because the price is above conventional speed bumps)

IDEA ASSESSMENT (MON) - SPEED BUMP

Market Opportunity: Non-Newtonian speed bump

CHALLENGE

IMPLEMENTATION OBSTACLES:

- Full-scale testing as the existing research only tested the concept with drop weights and bicycles at 5-25 km/h, not proper vehicles [16].
- Casing strength because it must withstand thousands of repeated axle loads without tearing or releasing fluid [17].
- The municipality would need to perform maintenance to check out things like the fluid condition.

Rating: High

TIME TO REVENUE:

Stage	Estimated Time
Develop full-size prototype	3–6 months
Car, truck and repeated-load testing	4–8 months
Weather and durability testing	6–12 months
Municipal approval	4–8 months
Production and Installation	1–2 months
First paid speed bump	18–30 months
Wider commercial sales	3–5 years

Rating: Super High

IDEA ASSESSMENT (MON) - SPEED BUMP

Market Opportunity: Non-Newtonian speed bump

EXTERNAL RISKS

- Municipal rejection as the design falls outside familiar speed-bump designs.
- Emergency-service opposition: In case of emergencies ambulances and police cars might still have a problem crossing.
- Weather could damage the fluid, especially in cold winter weather.
- There may be other speed bump ideas that offer the same exact purpose and are at a cheaper price.

Rating: High

Overall Potential => High

Overall Challenge => High

Ethical Considerations

ETHICAL CONSIDERATIONS

Three ideas, three ethical tensions

Social café

INCLUSION AND NUISANCE

Can a café for all neighbours exclude some and disturb others?

Security service

PRIVACY AND FAIR TREATMENT

Can a night patrol protect residents without watching or stigmatising some of them?

Non-Newtonian speed bump

ACCESSIBILITY AND HARM

Can a speed bump protect children without harming wheelchair users or emergency vehicles?

THE ETHICAL CYCLE (VAN DE POEL AND ROYAKKERS, 2007)

1 Problem statement | 2 Problem analysis | 3 Options for action | 4 Ethical judgment | 5 Reflection

Moral problem and course of action

MORAL PROBLEM STATEMENT

Under what conditions is a café for all neighbours acceptable if some residents cannot use it while those next door carry the noise?

Values in tension: equal access for everyone and quiet for neighbours

Why this could occur

Accessibility: Booking is online only, in Dutch and English, and a membership costs money.

Existing divides: Local social places are split by age, gender and background.

Nuisance: Evening events and open hours until 22:00 can disturb homes near the park.

Course of action

1. Make it easy to join: Allow phone booking, put printed calendars in shops and schools, add more languages and offer free or cheaper membership

2. Ask the neighbours. Agree on quiet hours and adjust the events schedule accordingly. Should run a pop-up trial first.

Standpoint. *If the pop-up reaches offline and low-income residents, build the café. If not, fix access/noticeability first.*

ETHICAL CONSIDERATIONS – SOCIAL CAFE

Stakeholders, interests and facts

STAKEHOLDER	INTEREST	KNOWN FACTS	UNKNOWN FACTS
Café visitors (all ages)	A nearby, affordable place to meet neighbours.	All six testers would support it, and residents say nothing is organised locally.	Whether people who do not already socialise will actually come.
Residents who could be left out	Take part without a smartphone, money or fluent Dutch.	A website prototype was shown, in Dutch and English; membership is €5 per month.	How many residents are offline or cannot pay, and who they are.
Neighbours next to the park	Quiet evenings and a clean, safe park.	Events and opening until 22:00 are planned; noise is a listed risk (MON).	Which homes are close enough to be affected and how complaints are handled.
Operator and municipality	A permitted, viable café that serves the community.	Permits are needed. the modest scenario earns about 370 euros a month.	Who runs it, who owns the site and who covers losses.

ETHICAL CONSIDERATIONS – SECURITY SERVICE

Moral problem and course of action

MORAL PROBLEM STATEMENT

Under what conditions is a paid night patrol acceptable if it makes residents feel safer but may watch or stigmatise some of them?

Values in tension: safety and peace of mind, and privacy and fair treatment

Why this could occur

Unclear mandate: Who guards answer to and what they may do is not defined yet.

Profiling: Young people who gather outside due to lack of social places can be seen as a problem.

Data and privacy: Reports and filming create information about residents, and there are no rules for it yet.

Course of action

1. Set clear rules first: Publish clear rules before launch about what guards may do, whether filming is allowed, and how residents can complain.

2. Ask the residents: Form a resident panel that includes young people, and hire professional guards.

Standpoint. *If young residents say they feel safer and not watched, launch with the rules. If not, tighten rules and start with guards on foot patrol and better lighting.*

ETHICAL CONSIDERATIONS – SECURITY SERVICE

Stakeholders, interests and facts

STAKEHOLDER	INTEREST	KNOWN FACTS	UNKNOWN FACTS
Residents of all ages	Feel safe and move freely after dark.	People feel least safe in the evening. All six testers felt more at ease.	Whether patrols cut real incidents (165 offences recorded in 2025).
Youth and minority groups	Use parks and streets without being treated as suspects.	Young people have few places to meet; one 16-year-old carries a knife to feel safe.	How would they feel with the guards around the neighborhood? Safe or watched?
Guards and security company	A clear mandate, safe working conditions, a steady contract.	The concept assumes two stewards for six hours each evening (MON).	Training, legal powers, liability and cooperation with the police.
Municipality and police	Fewer incidents and complaints, good value for money.	Likely payers: about €150,000 per neighbourhood per year (MON assumption).	Who is accountable, who owns the data and how information is shared.

ETHICAL CONSIDERATIONS – SPEED BUMP

Moral problem and course of action

MORAL PROBLEM STATEMENT

Under what conditions is a speed bump that hardens at speed acceptable if it protects children but may harm wheelchair users or delay ambulances?

Values in tension: safety for children and cyclists, and access and fairness for everyone

Why this could occur

Untested behaviour: How hard it gets, and its effect on bikes and wheelchairs, is unknown.

Emergency vehicles: Ambulances or police cars may need to cross fast and meet its hard state.

Unintended effects: We do not know how loud it is, how long it lasts or what it costs.

Course of action

1. Make crossing safe in other ways: Add a marked low-curb crossing, signs and lighting, so safety does not depend on the bump alone.

2. Ask the residents/people passing by: Test it on one street near a school with wheelchairs, bikes and emergency vehicles and measure speeds.

Standpoint. *If wheelchairs and emergency vehicles cross safely, use it near schools and parks. If not, use the safe crossing without the bump.*

ETHICAL CONSIDERATIONS – SPEED BUMP

Stakeholders, interests and facts

STAKEHOLDER	INTEREST	KNOWN FACTS	UNKNOWN FACTS
Pedestrians and cyclists	Safe crossings near the park, school and supermarket.	16 traffic incidents were recorded in 2025; all six testers would support it.	How much it really lowers speeds and prevents crashes.
People with limited mobility	Smooth, trip-free crossings with a wheelchair, rollator or cane.	Testers asked about tripping and wheelchair or rollator use; no tests exist yet.	How hard the bump gets and whether a wheelchair can cross safely.
Drivers and emergency services	Smooth passage at legal speed and fast response in emergencies.	In the concept the bump stays soft for slow drivers and hardens at high speed.	How ambulances and fire engines are affected, and at what speed it hardens.
Municipality and maintenance	Safer streets within budget, with little upkeep.	So far only AI-made images exist; no real test or pilot has been done.	Cost, durability, and upkeep of the oobleck chamber.

ETHICAL CONSIDERATIONS SUMMARY

Ethical comparison of the three ideas

IDEA	CORE TENSION	BIGGEST UNKNOWN	FIRST SAFEGUARD	CONCERN
Social café	Access and noise	Who is left out, and how many	Phone booking, more languages, house rules	**Mid**
Security service	Privacy and fair treatment	Does it reduce crime without unfair suspicion?	Published rules and a complaint route	**High**
Non-Newtonian speed bump	Safety and access	Is it safe for wheelchairs and emergency vehicles?	A real test with those users first	**Mid, until tested**

No idea is ruled out on ethical grounds, but each needs its safeguard built in before roll-out.

Final Idea Selection

FINAL IDEA SELECTION

Comparative overview of the three ideas

IDEA	DESIRABILITY (UEQ-S)	FEASIBILITY (MON)	VIABILITY (MON)	ETHICS
Social café	+0.96 UEQ-S; 6/6 would support	High potential; pop-up in weeks	Mid; thin margin, scales up	**Mid**
Security service	+0.56 UEQ-S (lowest); 5/6 support	High potential; one customer only	High; budget-dependent	**High**
Non-Newtonian speed bump	+1.15 UEQ-S (highest); 6/6 support	High long-term; untested, slow	Mid; pricier, unproven	**Mid, until tested**

All three tested well – the gap is speed, safety, and fairness.

FINAL IDEA SELECTION

Verdict and final decision

IDEA	DESIRABILITY VERDICT	FEASIBILITY & VIABILITY	ETHICS VERDICT	DECISION
Social café	Broadest appeal across all ages	Lowest barrier to entry	Manageable with simple fixes	**Selected**
Security service	Least wanted overall	Needs municipal funding	Risk of profiling youth	**Not selected**
Non-Newtonian speed bump	Highest score, but untested	Longest path to revenue	Unjudged until tested	**Parked**

Winner: Social Café – desirable, fast to pilot, lowest ethical cost.

Appendix

References

[1] AlleCijfers.nl. (2026). *Buurt Doornakkers-West in cijfers en grafieken*. Retrieved October 7, 2026, https://allecijfers.nl/buurt/doornakkers-west-eindhoven/
[2] AlleCijfers.nl. (2026). *Wijk Doornakkers in cijfers en grafieken*. Retrieved October 7, 2026, from https://allecijfers.nl/wijk/doornakkers-eindhoven/
[3] Gemeente Eindhoven. (2025). *Omgevingsprogramma stedelijke en maatschappelijke voorzieningen. Gemeenteblad 2025, 531379*.
https://zoek.officielebekendmakingen.nl/gmb-2025-531379.html
[4] Centraal Bureau voor de Statistiek. (2026, September 1). *Omzet horeca groeit in tweede kwartaal 2026 met ruim 2 procent*.
https://www.cbs.nl/nl-nl/nieuws/2026/36/omzet-horeca-groeit-in-tweede-kwartaal-2026-met-ruim-2-procent
[5] Nederlandse Veiligheidsbranche. (2026, May 20). *Branchescan beveiligingssector 2025: Meer beveiligers in vaste dienst; loonstijgingen remmen sectorale groei*.
https://www.veiligheidsbranche.nl/branchescan-beveiligingssector-2025-meer-beveiligers-in-vaste-dienst-loonstijgingen-remmen-sectorale-groei/
[6] Gemeente Eindhoven. (n.d.). *Subsidie gebiedsgericht werken*. https://www.eindhoven.nl/bestuur-en-beleid/subsidies/subsidie-gebiedsgericht-werken
[7] Justis. (n.d.). *Register vergunninghouders Wpbr*. Retrieved October 7, 2026, from https://justis.nl/registers/wpbr-register
[8] Centrum voor Criminaliteitspreventie en Veiligheid. (n.d.). *Stap 2: Praktijk vormgeven voor boa's*. Retrieved October 7, 2026, from
https://hetccv.nl/themas/veilige-woon-leefomgeving/buitengewoon-opsporings-ambtenaar-boa/stappenplan-gemeentelijke-handhavers/stap-2-vormgeven-praktijk/
[9] PIANOo. (n.d.). *Nationale procedures*. Retrieved October 7, 2026, from https://www.pianoo.nl/nl/inkoopproces/aanbestedingsprocedures/nationale-procedures
[10] Politie. (n.d.). *Aanvraagprocedure van een toestemming en/of legitimatiebewijs*. Retrieved October 7, 2026, from
https://www.politie.nl/informatie/aanvraagprocedure-van-een-toestemming-en-of-legitimatiebewijs.html
[11] Onderzoek040. (n.d.). *BuurtKijker: Wonen en woonomgeving*. https://onderzoek040.nl/mosaic/buurtkijker/wonen-en-woonomgeving
[12] Gemeente Eindhoven. (n.d.). *Snelheidsverlaging binnen de bebouwde kom*.
https://www.eindhoven.nl/stad-en-wonen/verkeer/verkeersveiligheid/snelheidsverlaging-binnen-de-bebouwde-kom
[13] Jung, I., Kim, E., Yu, W.-R., & Na, W. (2022). Development of smart speed bump using non-Newtonian fluid. *Composites Research, 35*(4), 277–282.
https://doi.org/10.7234/composres.2022.35.4.277
[14] Gemeente Eindhoven. (n.d.). *Snelheidsverlaging binnen de bebouwde kom*.
https://www.eindhoven.nl/stad-en-wonen/verkeer/verkeersveiligheid/snelheidsverlaging-binnen-de-bebouwde-kom
[15] POL Heteren. (n.d.). *Modular 10–4000 × 3000 mm, met zebrapad, stoep tot stoep*. Retrieved October 7, 2026, from
https://www.pol.nl/modular-10-4000x3000-mm-met-zebrapad-stoep-tot-stoep.html

References

[16] CROW. (2014). *Richtlijnen voor de toepassing van verkeersdrempels en -plateaus*. https://kennisbank.crow.nl/public/gastgebruiker/WOBU/Richtlijn_drempels%2C_plateaus_en_uitritten/Richtlijnen_voor_de_toepassing_van_verkeersdrempels_en_-plateaus/27209

[17] Stichting Brabant Maatjes. (2025, November 4). Social Café Tilburg – Ontmoet nieuwe mensen in een warme en creatieve setting [Photograph]. Retrieved October 7, 2026, from https://brabantmaatjes.nl/social-cafe-tilburg/

[18] Beauchamp, L. (2021, January 31). How to improve neighborhood security [Photograph]. Global Security Solutions. Retrieved October 7, 2026, from https://globalsecuritysolutionsllc.com/improving-neighborhood-security/

Interview guide (same 10 questions per prototype; Q6-Q7 are idea-specific)

1. In your own words, what is this idea and what is it meant to do?
2. Does it address a problem you actually experience here? Which one?
3. How useful would it be for you, especially in the evening?
4. What do you like most, and what is unclear or missing (in the idea or in how it is shown)?
5. Who would benefit most, and could it be a problem for anyone?
6. Idea-specific question 1
7. Idea-specific question 2
8. What one change would make it more useful?
9. Would you support it being introduced here?
10. Anything else you want to add?

Participant 1

1 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. At first it looks like a normal black-and-yellow speed bump. After you explained: the white stuff inside is a special material that is soft for slow cars and goes hard when a car is fast. **Q2.** Yes, traffic near the park and the main road matters to me. I cross there to get to the football pitch. **Q3.** Pretty useful, also in the evening because drivers go even faster when it's empty. **Q4.** I like that the street picture looks like our area and the cutaway shows what's inside. But nothing shows the hard part. I can't tell how hard it gets or if it would hurt a scooter or bike. Also I see it's an AI picture, so is it real? **Q5.** Kids and older people. I think it's a problem for delivery riders if it's too hard. **Q6.** *(Would slowing drivers make walking near parks feel safer?)* A bit, yes. In the evening other things matter to me more than traffic. **Q7.** *(Worries: noise, cost, maintenance, cyclists, emergency vehicles?)* Does it still work in winter and rain? And an ambulance would need to go over it. **Q8.** Show a video of a car and a bike going over it, slow and fast. **Q9.** Yes, support it. **Q10.** It's about traffic. For the evening I would like other measures as well.

1 – Prototype B: Security service (poster)

Q1. A night patrol with marked cars and guards, every night from 19:00 to 06:00. You can call or scan a code to report something. **Q2.** Yes, the evening. I would like to feel more at ease outside after dark, for example when I run. **Q3.** Could be useful, I would maybe run later if I knew they were around. But a car can't be everywhere. **Q4.** "Safe after dark" is catchy and the hours are easy to see. The phone number and QR code are empty, so I wouldn't know what to do. And it's unclear how the guards would treat the people they meet. **Q5.** Everyone who goes out at night. Some people might feel watched, which could be a problem. **Q6.** *(Would a night patrol make you more willing to be outside after dark?)* Yes, a little, if I see it often. If it comes once an hour it won't change much. **Q7.** *(How do you feel about patrols, privacy, trust?)* I don't mind guards. I like the line about knowing when to call the police. I don't want cameras on everything, and I'd want to know who they are. **Q8.** Fill in the number and QR, and show a guard talking to people on foot near the park, not only a car. **Q9.** Yes, if they are friendly and not aggressive. **Q10.** People should trust them. Otherwise nobody reports anything.

1 – Prototype C: Social café (desktop website UI)

Q1. A website for a big community café in the park. You see what's on, book free spots, check the menu, and anyone can just drop in. **Q2.** I would like more places to meet nearby. At the moment I usually go to the centre. **Q3.** Quite useful, because it's open until 22:00 and 23:00 on weekends. Board Game Club, Music Jam and Open Mic are in the evening. **Q4.** I like the calendar and that it's free and the prices are low. Unclear: it's a big desktop layout and I only use my phone, so is there a mobile version? And I see nothing for football or sports, or for people my age. It's mostly clubs for older people and families. **Q5.** Parents, older people, newcomers. Teenagers might feel it's not for them. **Q6.** *(Would you use it and go to the café? What would make you come back?)* Yes, if I can book fast on my phone and there's something like a match night or the Music Jam. The reminder before an event is useful. **Q7.** *(Would it help you meet neighbours, or mainly people who already know each other?)* It could help. "Make your own event" means I could start a football-screen night with friends, but I'd need to know it's also meant for us. **Q8.** Add a youth and sports category, and make the site work well on a phone. **Q9.** Yes. **Q10.** Where is it exactly? The address still says "[Street + number]". And I'd want to know the route there is well lit in the evening.

Participant 2

2 – Prototype B: Security service (poster)

Q1. Night patrol with marked cars and guards, every night, and you can call or scan to report something. **Q2.** Yes, I would like to feel more at ease in the evening, for example in the parks. **Q3.** Very useful. I'm rarely out at night, and I'd go out more if I knew someone was watching the parks. **Q4.** The poster is clear and I'd notice it at school or on a lamppost. I like "for every resident" and the 112 reminder. Missing: the phone number and QR code are placeholders, and I don't know who pays or how they work with the police. **Q5.** Families and older people. A problem could be if some people feel watched or treated unfairly. **Q6.** *(Would patrols make you more willing to be outside after dark?)* Yes, walking to friends or the garden after dinner. **Q7.** *(Feelings about surveillance, privacy, trust?)* It's fine if they only patrol public places. I'd want clear rules and a contact person in the neighbourhood. **Q8.** Add more languages. Language is a barrier for some neighbours, and it says only Dutch and English. **Q9.** Yes, I would support it. **Q10.** It works best together with places where people meet.

2 – Prototype C: Social café (desktop website UI)

Q1. A website for a café and community centre in the park, with activities, clubs, a noticeboard and a menu. You don't need a membership, you can just come. **Q2.** Yes, I would like a place nearby to meet friends. Now I usually go to the centre. **Q3.** Very useful. I could book Kids Lego Lab, Family Board Games or Homework Help for my kids, and Garden Club for me. And it's open until 22:00. **Q4.** I like that booking takes a minute, and the noticeboard, the volunteer roles and "pay what you can". What worries me: it's only in Dutch and English, and many neighbours have language barriers or don't use websites much. I also can't really see the indoor kids' playground. **Q5.** Parents, older people, newcomers. A problem could be noise for nearby houses, and people who are not online get left out. **Q6.** *(Would you use it and go to the café?)* Yes, weekly. Cheap coffee, the Dutch Conversation Café, and activities for the kids. **Q7.** *(Would it help meet neighbours?)* Yes, because there are organised things like the Neighbourhood Dinner. "Buy a coffee for a neighbour" is a nice idea. People don't just start talking in a café. **Q8.** Add more languages, print the weekly calendar for the school and shops, and show the kids' area clearly. **Q9.** Yes, definitely. **Q10.** Make sure the route there is pleasant and well lit in the evening. And who moderates the noticeboard?

2 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. A smart speed bump, soft for slow cars and hard for fast ones. From the pictures alone I wouldn't know, it looks like a regular bump. **Q2.** Yes. Traffic near the school and the park matters to me as a parent. **Q3.** Useful, in the morning and afternoon more than evening, but also in the evening. **Q4.** The street picture helps, I can imagine it by the park or school. I like that it protects children without hurting slow drivers. The pictures don't show the hard state, and I don't know if it has been tested anywhere. **Q5.** Children, and older people crossing. Maybe bad for bikes or people with a wheelchair. **Q6.** *(Safer near parks and schools?)* Yes, I'd feel safer letting the kids walk. **Q7.** *(Worries?)* Maintenance and cost, and winter weather. Is it durable? **Q8.** Show an example from another city or a test result, and combine it with a clearer crossing and better lighting. **Q9.** Yes. **Q10.** Prefer it near schools first.

Participant 3

3 – Prototype C: Social café (desktop website UI)

Q1. A website for a big café in the park where neighbours meet. There are clubs, dinners, a menu and a calendar, and everyone is welcome. **Q2.** Yes. A place in the park to sit and have tea would be nice, especially for older people. **Q3.** Useful in the afternoon. Chair Yoga for 60+, the Coffee Morning and the Walking Group are for people like me. In the evening I don't go out much. **Q4.** I like the tea at €1,80, the baklava and the halal labels. Someone thought about different tastes, and the prices are good. But it's a website with small text, only in Dutch and English, and I would need my grandson to help. The photos also show few older people like me. **Q5.** Older people and children. A problem if you need the internet to join, or if it becomes mostly for young families. **Q6.** *(Would you use it and go?)* I would go if tea is cheap and there are quiet tables for chess or cards. I saw the chess vote, I would vote. I would not use the website much, I would rather call the number or just walk in. **Q7.** *(Would it help meet neighbours?)* A bit. "Help with letters and forms" from Yusuf is very useful for many older people. Meeting through an activity works better than just sitting. **Q8.** Print the calendar for the local shops, in more languages and with bigger letters. **Q9.** Yes, I would support it. **Q10.** The shops are our meeting place now. The café should work together with them, not against them. And the address and phone number are still empty.

3 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. A speed bump that gets hard when a car goes fast. In the picture it looks like the normal black and yellow ones. I understood after you told me about the white material. **Q2.** Yes, calmer traffic at crossings would help me, because I walk every day. **Q3.** Very useful, because I walk every day and cross at the same places. **Q4.** I like that slow cars have no problem. The sketch with the crossing is clear. From a picture I can't feel it, I'd like to see a real one and step on it. I don't know if it is safe for old people on a bike or mobility scooter. **Q5.** Older people and children benefit. Drivers will complain. **Q6.** *(Safer near parks and schools?)* Yes, the crossing near the supermarket is where I would start. **Q7.** *(Worries?)* Noise, and it needs to look clearly like a bump. People should see it. **Q8.** Put clear paint or signs, and also check the traffic-light timing at crossings. **Q9.** Yes. **Q10.** A quick start would be good.

3 – Prototype B: Security service (poster)

Q1. Guards in marked cars every night, so people feel safer and can call them. **Q2.** For me the evenings are mostly calm, but others may feel differently. **Q3.** Not very useful for me, because I'm home in the evening. Maybe useful for others. **Q4.** A poster is simple, anyone can read it, better than something on a phone. I like that guards know when to call the police. But it's only in Dutch and English, and the number is missing. I worry that some people could feel watched or treated unfairly. The company name is missing too. **Q5.** Families and women walking alone. A problem for people who just want to meet outside and feel watched. **Q6.** *(Would patrols make you more willing to be outside after dark?)* A little. I'd go for a late walk if I saw them. **Q7.** *(Feelings about surveillance, privacy, trust?)* I trust it if the guards know the neighbourhood and talk to people. I don't trust a stranger company driving through. **Q8.** Hire people from the neighbourhood, add more languages, and hang the poster in the local shops where people go. **Q9.** Yes, but not before the other two. **Q10.** Safety also comes from people knowing each other.

Participant 4

4 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. A speed bump with a material inside that reacts to speed. Flat for slow, hard for fast. The cutaway shows a white filling inside. **Q2.** Yes. The crossing near the supermarket is busy, and I cycle a lot, so calmer traffic would help. **Q3.** Useful all day, especially when it is dark and I cycle home. **Q4.** The render looks professional and the street image makes it believable. But I only see one state. I'd like to see the fast-car case and how a bike feels on it. Also the bump looks quite low and narrow on the road, so can a car just avoid it? **Q5.** Pedestrians, cyclists and kids. Might be annoying for buses or drivers. **Q6.** *(Safer near parks and schools?)* Yes. But the crossing near the supermarket needs more than a bump. **Q7.** *(Worries?)* Cost and maintenance, and whether a municipality would invest in a new material. **Q8.** Test it on one street first and measure speeds, and show before and after. **Q9.** Yes. **Q10.** Add better lighting to secondary streets.

4 – Prototype C: Social café (desktop website UI)

Q1. A website for a community café and centre. There's a calendar, booking, clubs, a noticeboard, a menu, and a way to propose your own event. **Q2.** Yes. I mostly meet friends at campus or in the centre. I would like more ways to meet neighbours and find out what's going on. **Q3.** Useful in the evening, because it's open until 22:00. I'd go to the Dutch Conversation Café or the Language Swap, and maybe study there sometimes. **Q4.** I like the clear calendar, the booking with a reminder, the English switch, the "Welcome host" role and "Buy a coffee for a neighbour". The "Make your own event" steps are easy to follow. Unclear: there's no info about wifi or study space, the address is still a placeholder, and I can't see how crowded it will be. **Q5.** Families, older residents and students. A problem for people who aren't online, or if the kids get too noisy. **Q6.** *(Would you use it and go?)* Yes, once or twice a week. I'd use the reminder option, and I might propose a language or chess evening myself. **Q7.** *(Would it help meet neighbours?)* Yes. Clubs with a named host, like Board Game Club with Ahmed, make it less scary to join alone. **Q8.** Add wifi and study info, a map with the route, and a mobile version. **Q9.** Yes. **Q10.** Make it welcoming for internationals as well, and keep some offline info for people without internet.

4 – Prototype B: Security service (poster)

Q1. Guards in marked cars that patrol at night, and you can report problems by phone or QR code. **Q2.** Yes, in the evening I would like to feel comfortable cycling home, also along the canal and side streets. **Q3.** Useful. I'd feel more comfortable cycling home late. **Q4.** The poster is clear and the bilingual text is good for me. I like that the QR code shows tonight's patrol times. Unclear: what happens when I report something, how fast they respond, and how my privacy is handled. The number and QR code are placeholders. 19:00 to 06:00 is very long, is it really every night? **Q5.** Women and people walking alone. A problem for people who feel watched or unfairly treated. **Q6.** *(More willing to be outside after dark?)* Yes, along the main routes. I'd also like to see it along the canal. **Q7.** *(Surveillance, privacy, trust?)* I'd want clear rules: no filming without notice, and a complaint option. **Q8.** Fill in the QR code, add a map of patrol routes, and include routes along the canal and side streets. **Q9.** Yes, with clear rules. **Q10.** Patrol cars are visible but expensive. I wonder who pays.

Participant 5

5 – Prototype B: Security service (poster)

Q1. A night patrol with marked cars and uniformed guards, every night, and you can call or scan a code to report something. **Q2.** Yes. Some evenings I finish a shift around 22:00 and walk home. With my cane I walk slowly, so I would value feeling at ease on that route. **Q3.** Very useful for me, because it's exactly the time I'm outside. **Q4.** I like "for every resident, all ages and backgrounds" and that the guards are trained professionals. The letters are big, which is good. Unclear: how fast they come, whether they can meet me on my route, and the phone number is empty. It's in Dutch and English only. **Q5.** People who work evenings, older people, women alone. A problem if people feel watched just for being outside. **Q6.** *(Would a night patrol make you more willing to be outside after dark?)* Yes, I would worry less on my way home. I'd feel better if I saw the car on my street, not only in the park. **Q7.** *(Surveillance, privacy, trust?)* I'm fine with guards if they're polite and visible. I don't want cameras filming people's houses. I'd trust it more if some of them are from the neighbourhood. **Q8.** Add a way to call and have them meet or follow me home, and add more languages. **Q9.** Yes, I would support it. **Q10.** Good lighting and even paving on the route home would help as well.

5 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. It looks like a normal black-and-yellow bump. After you explained: the white filling is soft for slow cars and hardens for fast ones. **Q2.** Yes. With a cane I cross slowly, so calmer traffic at the crossing near the supermarket would help. **Q3.** Useful, in the day and in the evening when I come home. **Q4.** I like that it slows only the fast drivers and the street picture looks real. But for me a bump is also a tripping risk. The pictures don't show a ramp edge, a crossing, or a person walking on it. And the hard state isn't shown. **Q5.** Older people and children benefit. A problem for people with a rollator, wheelchair or bike if the surface is uneven. **Q6.** *(Safer near parks and schools?)* Yes, if cars slow down before the crossing. That is the moment that matters most for me. **Q7.** *(Worries?)* Tripping, ice and snow, noise at night, and who maintains it. **Q8.** Combine it with a marked, low-curb crossing, and show a person with a cane or rollator in the picture. **Q9.** Yes. **Q10.** Even paving on my street would help too, so it could be done at the same time.

5 – Prototype C: Social café (desktop website UI)

Q1. A website for a café and community centre. There are activities, clubs, a menu and a calendar, you can book a free spot, and everyone can come. **Q2.** I see my family and colleagues, but few neighbours. I would like a place nearby to sit down with a tea and meet people. **Q3.** Useful in the afternoon and some evenings. Chair Yoga for 60+ at €2 and the Coffee Morning look good, and it's open until 22:00. **Q4.** I like the mint tea at €2,40, the halal labels and "pay what you can". The accessibility box in the booking form and the FAQ about wheelchairs are good. Unclear: is there a step-free entrance, are there benches and an accessible toilet, and how do I get there from the south? The site is long and complicated on my phone, and I would rather call. **Q5.** Older people and people who need care or support. A problem for people who are not online or who don't read forms easily. **Q6.** *(Would you use it and go?)* Yes, with a friend, if there are comfortable chairs and not too far to walk. I might use the booking form once, but I would prefer a phone call. **Q7.** *(Would it help meet neighbours?)* Yes. "Welcome host" is important, I would not like to walk in alone the first time. "Help with letters and forms" and the walking-buddy notice are useful for people like me. **Q8.** Add photos of the entrance and a step-free route, a clear phone number for booking, and more languages. **Q9.** Yes. **Q10.** The address is still a placeholder. The café should be easy to reach from all parts of the neighbourhood, including the south.

Participant 6

6 – Prototype C: Social café (desktop website UI)

Q1. A website for a café and community centre in the park, with a calendar, booking, clubs, a menu, a noticeboard and a form to start your own event. It's open until 22:00. **Q2.** Yes, I would like a place nearby to go with my toddler and meet other parents. Now I usually go to the centre. **Q3.** Useful in the afternoon and some evenings. Kids Lego Lab and Family Board Games are relevant, although my child is still a bit young for them. **Q4.** I like the calendar, the booking with a reminder, free entry and "pay what you can". Unclear: is there anything for babies and toddlers, is it easy to get in with a pushchair, and is there a changing table? I can't see the indoor kids' area. It's also a desktop layout, and I would mostly look at it on my phone while out. **Q5.** Parents with young children, older people, newcomers. A problem for people who are not online, or noise for nearby homes. **Q6.** *(Would you use it and go?)* Yes, weekly. A quiet morning slot for parents with babies and a low-priced coffee would make me come back. **Q7.** *(Would it help meet neighbours?)* Yes, especially through organised things like the Coffee Morning, or a parents' meet-up I could set up with "Make your own event". A named host makes it easier to join alone. **Q8.** Add a mobile version, a baby and toddler category, and photos of the entrance and facilities. **Q9.** Yes. **Q10.** The address is still a placeholder. Short, simple information in Dutch and English would help everyone.

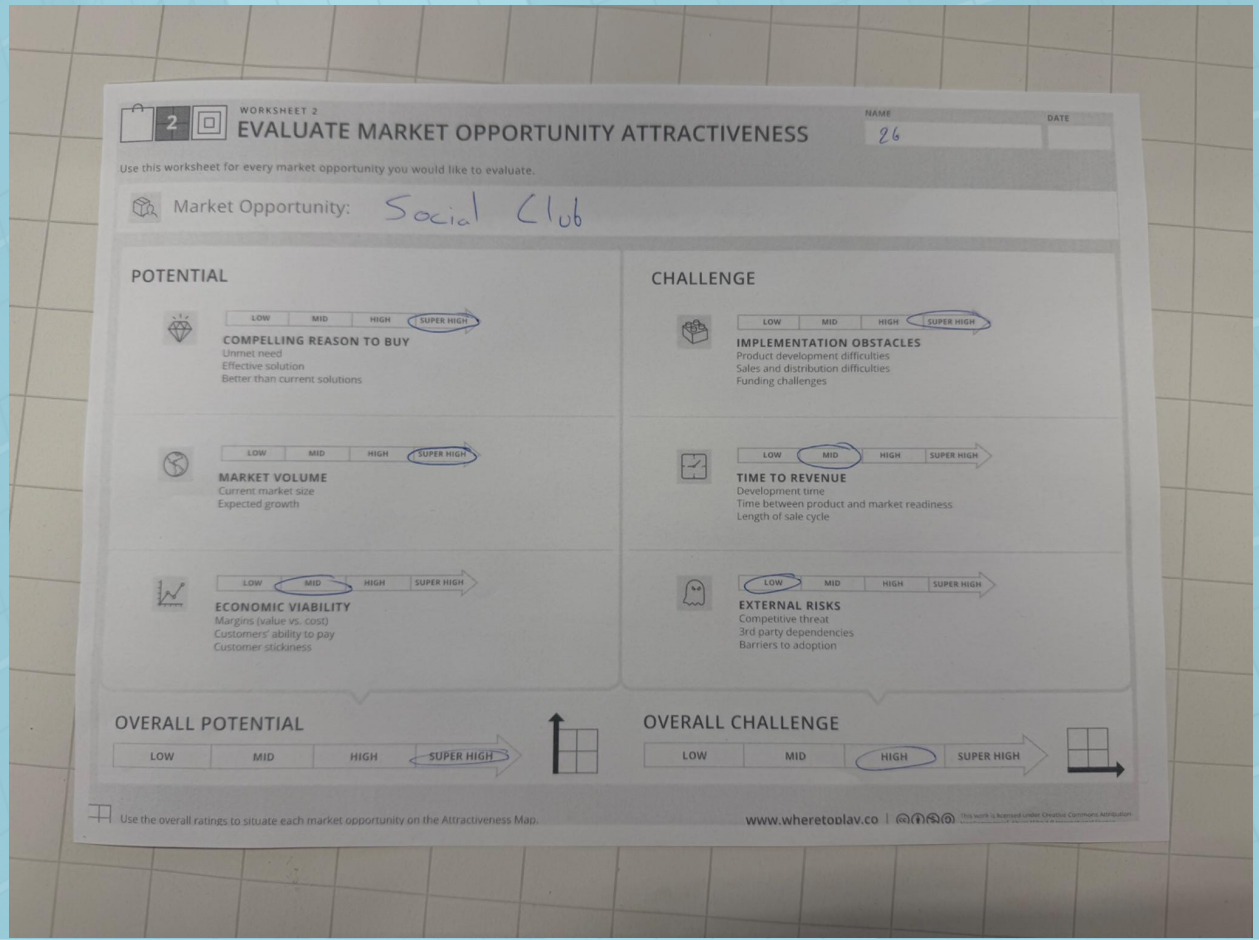
6 – Prototype B: Security service (poster)

Q1. A night patrol with marked cars and uniformed guards, every night from 19:00 to 06:00, and a phone number or QR code to report a concern. **Q2.** Partly. In the evening I'm usually at home with my toddler, but on office days I sometimes walk home late and I like feeling at ease on the route. **Q3.** Moderately useful, a few evenings a week. **Q4.** The headline and the hours are clear and the poster looks professional. Unclear: the phone number and QR code are placeholders, how quickly the guards respond, who runs the service, and how reports are handled. **Q5.** People who come home late, older people, parents. A problem if some residents feel watched, or if the patrol replaces other things like lighting and activities. **Q6.** *(Would a night patrol make you more willing to be outside after dark?)* Yes, a little, on the routes around the park, if I see the cars regularly. **Q7.** *(Surveillance, privacy, trust?)* I'm fine with guards in public places if the rules are clear: what they can do, how to complain, and no filming of homes. **Q8.** Add a QR code that works, a patrol map, a way to report without a phone call, and simple text in more languages. **Q9.** Yes, with clear rules. **Q10.** Start with a pilot of a few months and an evaluation, so residents can see whether it helps.

6 – Prototype A: Non-Newtonian speed bump (sketch + AI render + AI street image)

Q1. A speed bump that reacts to speed: soft for slow cars, hard for fast ones. In the pictures it looks like a normal bump, and after you explained it made sense. **Q2.** Yes, traffic near the park and on the routes to the school matters to me when I walk with my toddler in the pushchair. **Q3.** Useful mostly in the daytime, and also on weekend and evening walks. **Q4.** I like that it doesn't affect slow drivers, and the street image looks like a street I know. Unclear: how it feels with pushchair wheels, the hard state isn't shown, and whether it covers the whole crossing. **Q5.** Families with young children, older people, cyclists. A problem for pushchairs, wheelchairs and cargo bikes if the surface is rough or the edges are steep. **Q6.** *(Safer near parks and schools?)* Yes, if drivers slow down before the crossing. I would feel calmer crossing with the pushchair. **Q7.** *(Worries?)* Noise at night for nearby homes, cost and maintenance, winter weather, and how it copes with heavy delivery vans. **Q8.** Show a short video with a pushchair and a bike crossing it, and add a smooth ramp edge. **Q9.** Yes. **Q10.** Start next to the school and the playground, and combine it with a marked crossing.

Workshop Evidence (Market Opportunity Attractiveness for the Social Cafe)



WORKSHEET 2

EVALUATE MARKET OPPORTUNITY ATTRACTIVENESS

NAME 26 DATE

Use this worksheet for every market opportunity you would like to evaluate.

Market Opportunity: Social Club

POTENTIAL

COMPELLING REASON TO BUY — LOW | MID | HIGH | SUPER HIGH
Unmet need
Effective solution
Better than current solutions

MARKET VOLUME — LOW | MID | HIGH | SUPER HIGH
Current market size
Expected growth

ECONOMIC VIABILITY — LOW | MID | HIGH | SUPER HIGH
Margins (value vs. cost)
Customers' ability to pay
Customer stickiness

CHALLENGE

IMPLEMENTATION OBSTACLES — LOW | MID | HIGH | SUPER HIGH
Product development difficulties
Sales and distribution difficulties
Funding challenges

TIME TO REVENUE — LOW | MID | HIGH | SUPER HIGH
Development time
Time between product and market readiness
Length of sale cycle

EXTERNAL RISKS — LOW | MID | HIGH | SUPER HIGH
Competitive threat
3rd party dependencies
Barriers to adoption

OVERALL POTENTIAL

LOW | MID | HIGH | SUPER HIGH

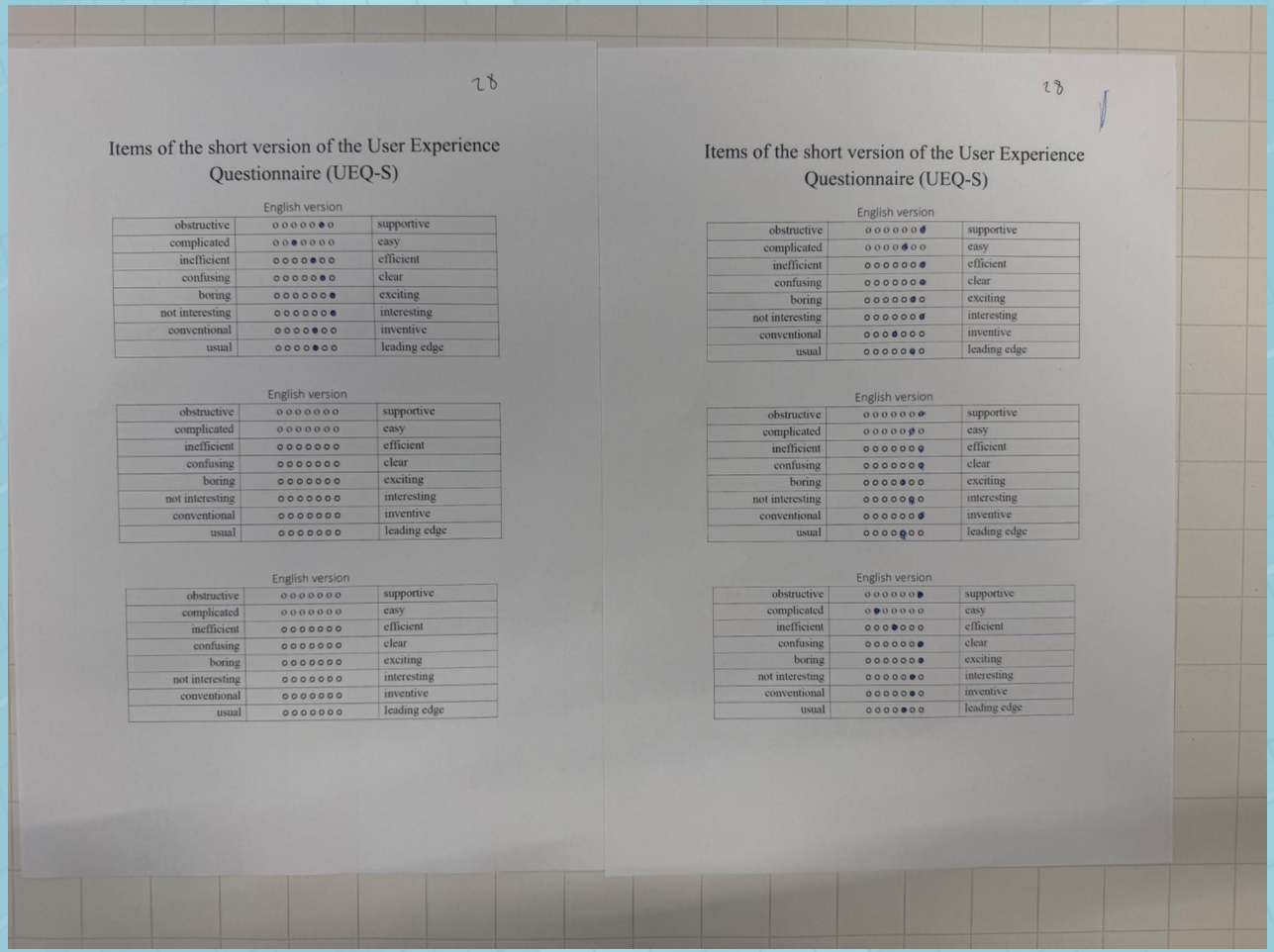
OVERALL CHALLENGE

LOW | MID | HIGH | SUPER HIGH

Use the overall ratings to situate each market opportunity on the Attractiveness Map.

www.wheretoplay.co | This work is licensed under Creative Commons Attribution

Workshop Evidence (UEQ-S from Group 28 for the Social Cafe)



28

Items of the short version of the User Experience Questionnaire (UEQ-S)

English version

obstructive	○○○○○●○	supportive
complicated	○○●○○○○	easy
inefficient	○○○○●○○	efficient
confusing	○○○○○●○	clear
boring	○○○○○○●	exciting
not interesting	○○○○○○●	interesting
conventional	○○○○●○○	inventive
usual	○○○○●○○	leading edge

English version

obstructive	○○○○○○○	supportive
complicated	○○○○○○○	easy
inefficient	○○○○○○○	efficient
confusing	○○○○○○○	clear
boring	○○○○○○○	exciting
not interesting	○○○○○○○	interesting
conventional	○○○○○○○	inventive
usual	○○○○○○○	leading edge

English version

obstructive	○○○○○○○	supportive
complicated	○○○○○○○	easy
inefficient	○○○○○○○	efficient
confusing	○○○○○○○	clear
boring	○○○○○○○	exciting
not interesting	○○○○○○○	interesting
conventional	○○○○○○○	inventive
usual	○○○○○○○	leading edge

28

Items of the short version of the User Experience Questionnaire (UEQ-S)

English version

obstructive	○○○○○○●	supportive
complicated	○○○○●○○	easy
inefficient	○○○○○○●	efficient
confusing	○○○○○○●	clear
boring	○○○○○●○	exciting
not interesting	○○○○○○●	interesting
conventional	○○○●○○○	inventive
usual	○○○○○●○	leading edge

English version

obstructive	○○○○○○●	supportive
complicated	○○○○○●○	easy
inefficient	○○○○○○●	efficient
confusing	○○○○○○●	clear
boring	○○○○●○○	exciting
not interesting	○○○○○●○	interesting
conventional	○○○○○○●	inventive
usual	○○○○●○○	leading edge

English version

obstructive	○○○○○○●	supportive
complicated	○●○○○○○	easy
inefficient	○○○●○○○	efficient
confusing	○○○○○○●	clear
boring	○○○○○○●	exciting
not interesting	○○○○○●○	interesting
conventional	○○○○○●○	inventive
usual	○○○○●○○	leading edge

Thank you, for your attention!